

WILLIAM TRAUB

Available for Co-op: July – December 2026

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Education

Northeastern University, Khoury College of Computer Science

September 2024 – Present

Candidate for Bachelor of Science in Computer Science and Mathematics

Boston, MA

- Dean's List, All Semesters (2024–2026) — Expected Graduation: May 2028
- Relevant Coursework: Object-Oriented Design, Data Structures & Algorithms, Theory of Computation, Advanced Linear Algebra, Probability & Statistics, Logic and Computation

Technical Skills

Languages: Python, Java, JavaScript, OCaml, Racket, C++

Data & ML: NumPy, Pandas, SciPy, Matplotlib, Machine Learning, Neural Networks, Anomaly Detection

Tools: Git, API Integration, Log Analysis, Scripting, Testing Automation, Agile Development, $\text{\LaTeX}$

Experience

Trackside Teen Center of Wilton

September 2021 – August 2024

Computer Science Teacher

Wilton, CT

- Taught foundational programming skills in Python and JavaScript to students ages 11–13.
- Led workshops and supervised projects to develop problem-solving and computational thinking.
- Mentored students in peer-tutoring roles to foster long-term engagement in STEM.

Projects

TerraNova | *Python, NumPy, Pandas, Matplotlib, Neural Networks*

February 2025

- Developed a machine learning application to assist city planners in identifying and classifying brownfield sites using geospatial and environmental features.
- Designed and trained a feedforward neural network on data from over 280 sites; tuned hyperparameters and regularization to improve generalization.
- Implemented data processing pipelines and optimized classification algorithms, improving accuracy by **18%**.

Digital Study Repository | $\text{\LaTeX}$, *Git*

June 2024

- Digitized academic notes using $\text{\LaTeX}$, building a structured archive spanning abstract algebra, real analysis, linear algebra, algorithms, and logic.
- Maintained an open-source GitHub repository used by students and independent learners; structured for long-term reference and readability.

Additional Experience

Oregon Programming Languages Summer School (OPLSS) | *Participant*

June 22 – July 4, 2026

- Selected for a competitive, graduate-level summer school on programming language theory, type theory, and formal verification.
- Engaged with coursework on type systems, semantics, and mechanized proof — directly relevant to compiler design, language tooling, and verified software development.

Directed Reading Program, Northeastern University | *Participant*

2024 – 2025

- Completed an independent reading of Aluffi's *Algebra: Chapter 0* with a PhD mentor, covering category theory, group theory, and homological algebra.

Clubs & Activities

Northeastern Math Club | *Event Planner*

2024 – Present

- Organized math talks, social events, and academic programming for club members.

Interests

Machine Learning, Formal Verification, Programming Language Theory, Game Development, Digital Art, Greek (proficient), Mandarin Chinese (intermediate), Hiking